EXPERIMENT 18

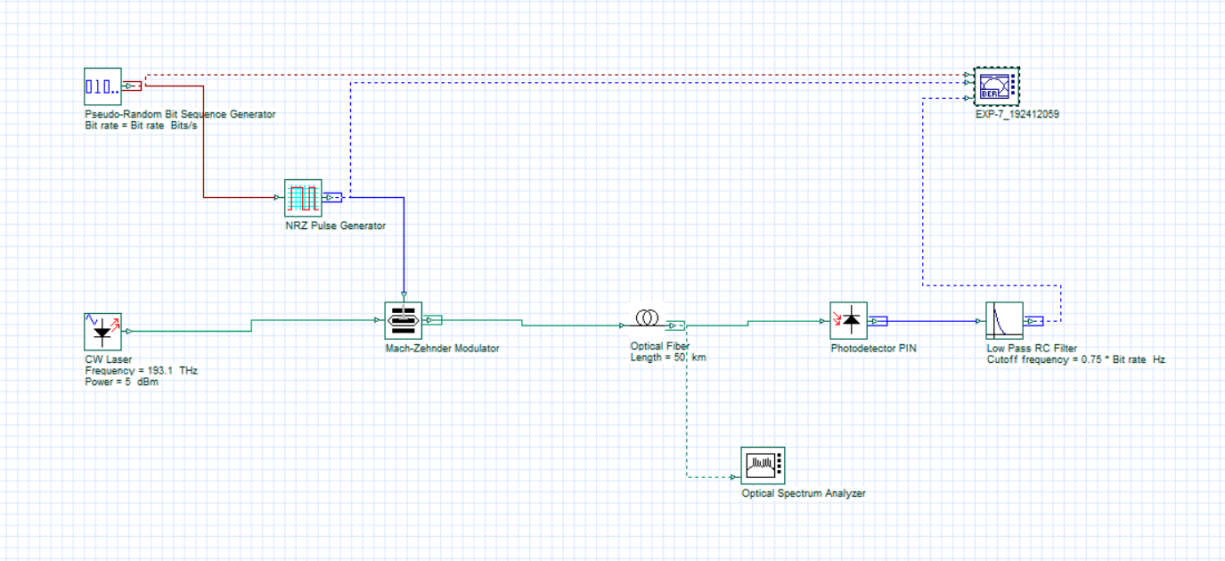
OPTICAL SPECTRUM AND OSNR ANALYSIS USING OPTISYSTEM

OBJECTIVES :

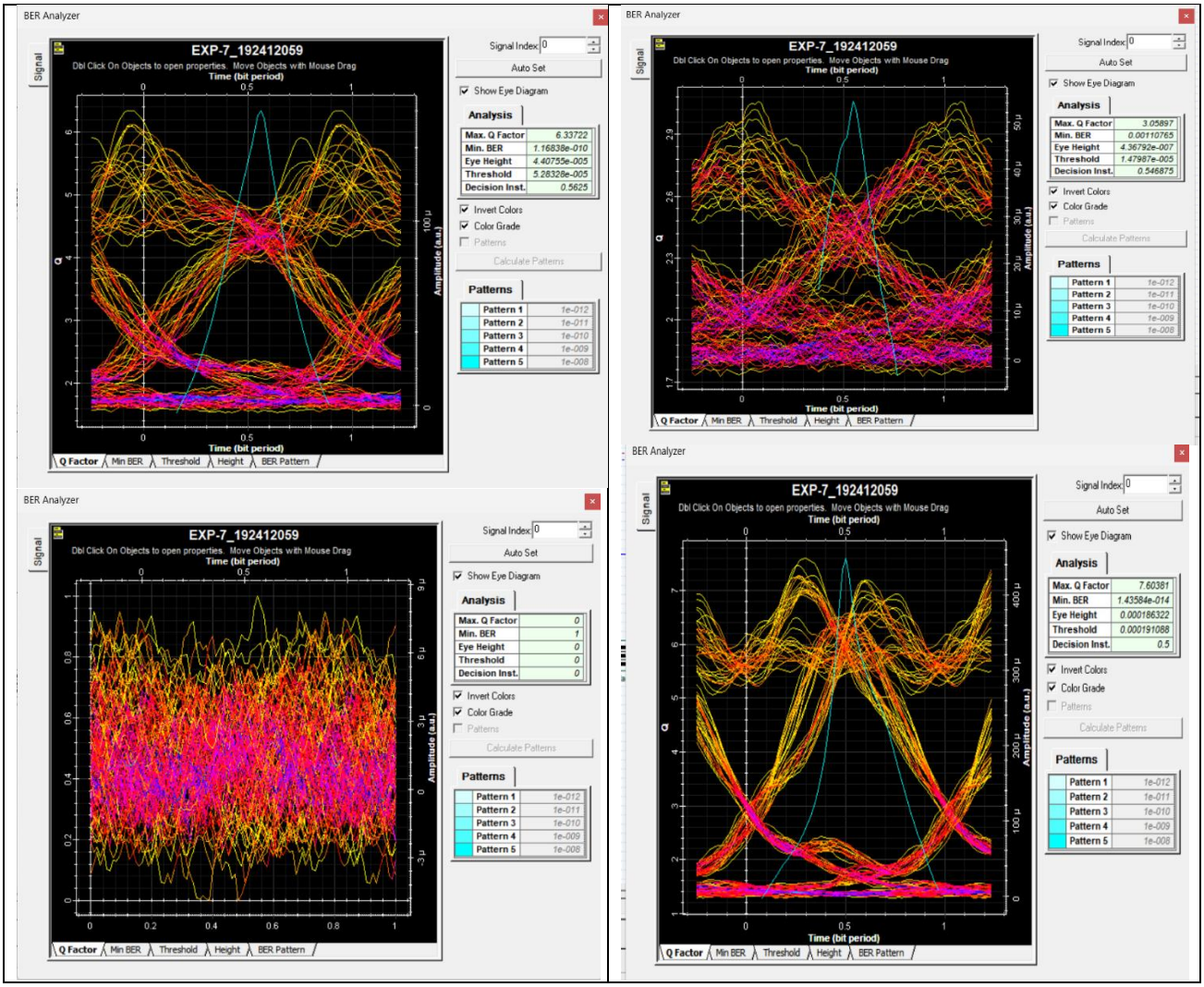
1. Design an optical communication system using OptiSystem.
2. Study how source power affects optical spectral intensity.
3. Analyze how fiber length affects attenuation, dispersion, and nonlinear effects.
4. Study the combined effect of source power and fiber length.

SOFTWARE REQUIRED: Opti System.

CIRCUIT DIAGRAM:



OUTPUTS:



REFLECTION:

1. Learned to design an optical communication system using OptiSystem.
2. Understood how to observe and analyse the optical spectrum.
3. Gained knowledge about OSNR and its importance.
4. Understood how noise affects optical signal quality.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & Simulation Setup	Correct selection and connection of OptiSystem components (CW Laser, MZM, PRBS, NRZ Pulse Generator, SMF, PIN Photodiode, Low Pass Filter, OSA, BER Analyzer).	All optical and electrical components are correctly selected, connected in proper sequence, and configured without errors.	All components are present with minor layout or connection errors, but the simulation link remains functional.	Component selection is partially complete; minor connection or configuration errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of physical input parameters: Wavelength (1550 nm), Attenuation (0.2 dB/km), Dispersion (16 ps/nm/km), Source Power, and Fiber Length across all test cases.	All input parameters and test cases (Test Cases 1, 2, and 3) are accurately set and executed per procedure.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but one or more test case parameter variations are missing.	Incorrect input parameter settings or failure to run the required test cases.	20%
Measurement & Data Collection	Accurate extraction and recording of Min BER, Max Q-factor, spectral power, and eye diagram displays.	Precise data extraction across all test cases, neatly tabulated with complete optical spectra and clear eye diagrams.	Data is collected with minor omissions in recording BER, Q-factor, or spectral readings.	Partial data collection with incomplete tables or missing eye diagram readings.	Inaccurate, incomplete, or missing data and measurements.	20%
Data Analysis & Spectral Interpretation	Detailed evaluation of source power impact, fiber length attenuation, dispersion, and non-linear distortion on link performance.	Comprehensive analysis correctly relating source power and length variations to spectral broadening, attenuation, dispersion, BER, and Q-factor degradation.	Clear analysis identifying basic trends in BER and Q-factor relative to power and distance variations.	Basic attempt at result analysis without explaining physical mechanisms like dispersion or non-linearity.	Erroneous or missing analysis of simulation results and graphs.	20%
Report & Conclusion	Logical deduction of optimal operating parameter range, visual presentation of output graphs, and structural clarity.	Report features clear graphs, complete tables, and insightful conclusions identifying optimal parameters for peak Q-factor and lowest BER.	Satisfactory report with relevant graphs and sound concluding summary.	Incomplete report with missing graphs, eye diagrams, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION :

Thus, the optical spectrum and OSNR analysis of the optical communication system was successfully performed using OptiSystem software. The optical spectrum analyser was used to observe the signal spectrum and measure the optical signal-to-noise ratio (OSNR). The analysis helped to understand the effect of noise on signal quality and the importance of maintaining a good OSNR for reliable optical communication. Hence, this experiment provided practical knowledge of optical signal quality and system performance..